

# *Air Regulations*

FAIR  
NOTES



# Introduction

## 1. Types of Flight

- **Dual Flight** - When you are flying with an instructor on board.
- **Solo Flight** - When you are alone (only soul) in the A/C (Aircraft) i.e. there is no other pilot / instructor / passenger (pax.)
- **PIC Flight** - 'Pilot in Command'. When you are the main pilot of the A/C. i.e. other than you, there are other pilots or pax. or both.

## 2. Licenses

- Student Pilot License.
- Private Pilot License.
- Commercial Pilot License.
- Airline Transport Pilot License.

\* NOTE - Conditions for CPL

UPTO 5700 kg - CPL holders can act as PIC.

ABOVE 5700 kg - Only ATPL can act as PIC.

### 3. License Specifications

| Licenses | Min Age | Max Age                               | Medical Certificate | Validity | Privileges                                      | Limitation                                              | Flying Hours Required |
|----------|---------|---------------------------------------|---------------------|----------|-------------------------------------------------|---------------------------------------------------------|-----------------------|
| SPL      | 16 y.o. | Till Medically fit                    | Class 2             | 5 years  | Can Fly Solo                                    | Cannot carry passengers or fly for hire.                | —                     |
| PPL      | 17 y.o. | Till Medically fit                    | Class 2             | 10 years | Can Carry Passengers                            | Cannot fly for hire.                                    | 40 hrs                |
| CPL      | 18 y.o. | 60 y.o. (or)<br>65 y.o. in conditions | Class 1             | 10 years | Can Fly for Hire (carry pax or cargo for money) | Cannot act as PIC of an A/C above 5700 kg or 12,500 lbs | 200 hrs               |
| ATPL     | 21 y.o. | 60 y.o. (or)<br>65 y.o. in conditions | Class 1             | 10 years | Can act as PIC of A/C above 5700 kgs            | —                                                       | 1500 hrs              |

#### 4. Medical Certificate Types -

| Medical \ Age | Less than 40 | Above 40  |
|---------------|--------------|-----------|
| Class 1       | 12 months    | 6 months  |
| Class 2       | 24 months    | 12 months |

#### 5. NATO Phonetic Alphabet

|              |              |
|--------------|--------------|
| A - Alpha    | N - November |
| B - Bravo    | O - Oscar    |
| C - Charlie  | P - Papa     |
| D - Delta    | Q - Quebec   |
| E - Echo     | R - Romeo    |
| F - Foxtrot  | S - Sierra   |
| G - Golf     | T - Tango    |
| H - Hotel    | U - Uniform  |
| I - India    | V - Victor   |
| J - Juliette | W - Whiskey  |
| K - Kilo     | X - X-Ray    |
| L - Lima     | Y - Yankee   |
| M - Mike     | Z - Zulu     |

JAVIN

Juliette - Alpha - Victor - India - November

## 6. Go or No Go Decision (IM SAFE Checklist)

- Illness
- Medication - Do not fly after taking any medicine that can cause drowsiness.
- Stress -
  - I. Acute (Short Term)
  - II. Chronic (Long Term)
- Alcohol - Bottle to throttle needs a minimum gap of 12 hours. ( $<0.04\%$ )
- Fatigue -
  - I. Acute (Short Term)
  - II. Chronic (Long Term)
- Eating - Always be well fed before a flight.

## 7. Cabin Crew Requirements

- Calculated according to the number of passenger seats on an A/C.
- No cabin crew is required below 10 seats for pax. in an A/C

| No. of Seats                     | Cabin Crew Req. |
|----------------------------------|-----------------|
| 10-49                            | 1               |
| 50-99                            | 2               |
| 100-149                          | 3               |
| 150-199                          | 4               |
| 200-249                          | 5               |
| and an additional 1 per 50 seats |                 |

## 8. First Aid Kit Requirements

- This requirement is calculated according to the number of passengers.

| No. of pax. | First Aid Kits |
|-------------|----------------|
| 0-100       | 1              |
| 101-199     | 2              |
| 200-299     | 3              |
| 300 & above | 4              |

## 9. About ICAO.

- ICAO - International Civil Aviation Organization.
- Formed at the Chicago Convention on Dec 7, 1944.
- Originally 52 members.
- Has 190+ members today
- HQ at Montreal, Quebec, Canada.



### - Indian Hierarchy

Ministry of Civil Aviation (MoCA)  
(Reports to ICAO)



**DGCA**

- Responsible for licensing & certification. They set standards for aerodromes & audits them each year.

**AAI**

- They are responsible for maintaining the aerodromes & airspaces (unless privatized ex. GMR airport, Delhi.)

## 10. Documents that need to be carried in an A/C

### I. Pilot's Documents (PPM) -

- Pilot's License
- Photo Identification (ex. Passport)
- Medical Certificate.

### II. Aircraft's Documents (AROW) -

- Airworthiness Certificate
  - Fit to Fly Certificate
  - Issued by DGCA

| A/C Age        | Validity |
|----------------|----------|
| 0-10 yrs       | 5 yrs    |
| 10-20 yrs      | 2 yrs    |
| 20 yrs & above | 1 yr.    |

- Registration Certificate
  - To see the current owner & their address
  - Issued by MoCA.
  - Records maintained by DGCA.
  - Lifelong Validity.
- Operating Limitations - Pilot's Operating Handbook (POH)
  - Contains all information about an A/C and its systems
- Weight & Balance

# Types of an Aircraft

• **Aircraft** - An aircraft is anything which is capable to fly in the air whether with the help of an engine or without the help of an engine.

• Categories of an Aircraft -

## I Airplane -

- Capable of flying in the air.
- Engine Driven
- Fixed Wings Fuselage.



### \* NOTE -

- Smoking is only permitted on an A/C when mentioned in the COA. (Certificate of Airworthiness)
- A flight engineer is only required on an A/C when specifically mentioned in the A/C's certificate of airworthiness.

## II Helicopter -

- Capable of flying in the air.
- Engine Driven
- Rotor Blades



## III Glider -

- Capable of flying in the air.
- Non engine driven.
- Fixed Wings



## IV Hot Air Balloon -

- Capable of flying in the air.
- Non Engine driven.



## ◦ Aircraft in an Emergency -

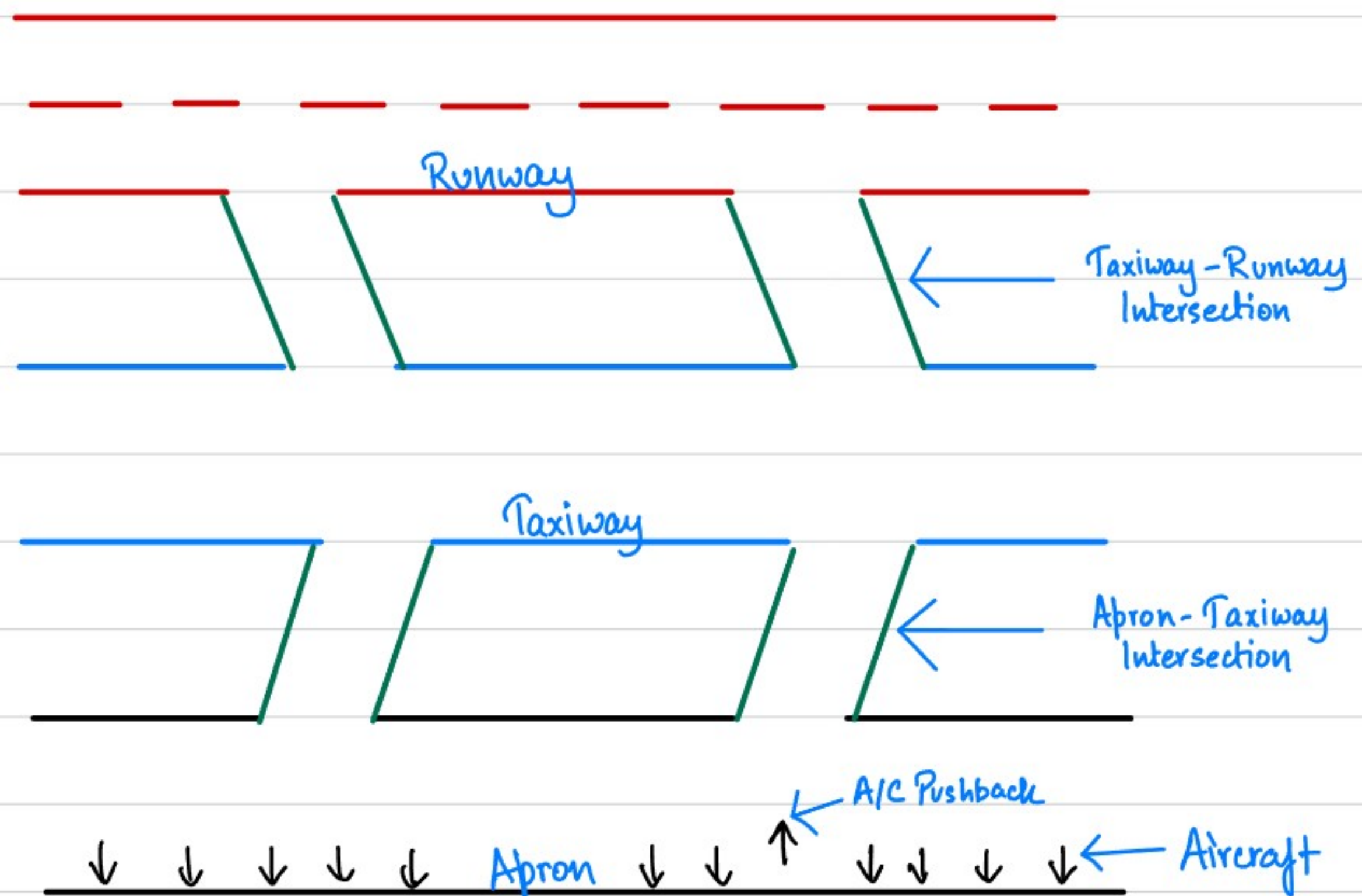
MAYDAY - MAYDAY - MAYDAY

- This means the A/C needs imminent help.

PAN PAN - PAN PAN - PAN PAN

- This means, if not paid attention, situation could turn into an emergency.

## ◦ Important Parts of an Aerodrome.



◦ **Taxi** - The movement of an A/C under its own power on ground.

◦ **Apron** - Where A/C's are parked at an airport.

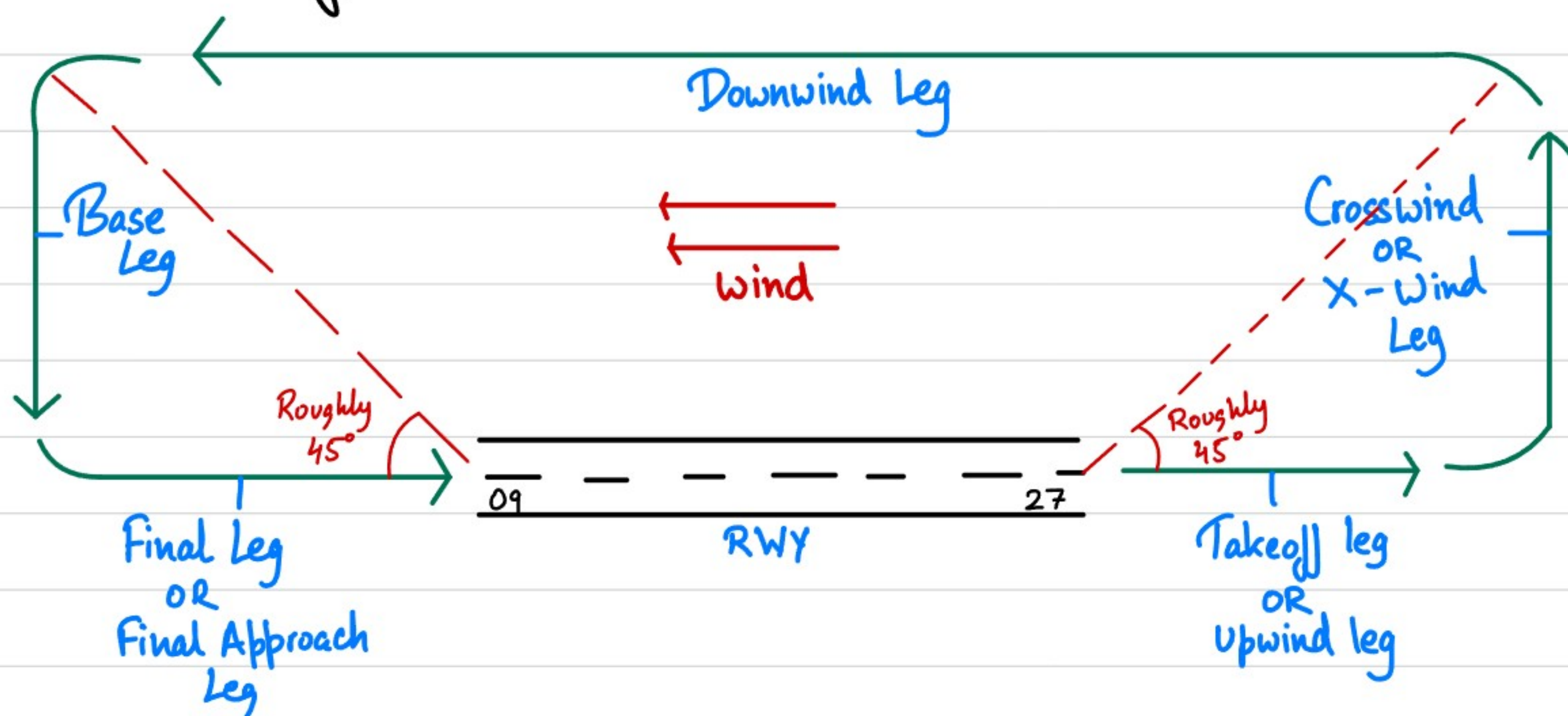
◦ **Taxiway** - Used to taxi from apron to runway & vice versa under its own power.

◦ **Runway** - Used for take off & landing

◦ **Movement Area** - Apron (+) TWY (+) RWY

◦ **Maneuvering Area** - TWY (+) RWY

## • Standard Traffic Pattern -



A standard fixed wing traffic pattern consists of left turns, has 5 designated legs, and is flown at 1000 ft. AGL (Above Ground Level).

1. Upwind Leg OR Takeoff Leg.
2. Crosswind OR X-Wind Leg.
3. Downwind Leg.
4. Base Leg.
5. Final Leg OR Final Approach Leg.

A runway will have a non-standard traffic pattern for ex. - a right pattern due to terrain or obstacles. This will be mentioned in the AIP.

An AIP (Aeronautical Information Publication) of a country describes all details of that country's airports. A pilot must study an airport before they fly to it.

A traffic pattern is generally followed by planes following VFR (i.e. Visual Flight Rules).

# Nationality & Registration Markings

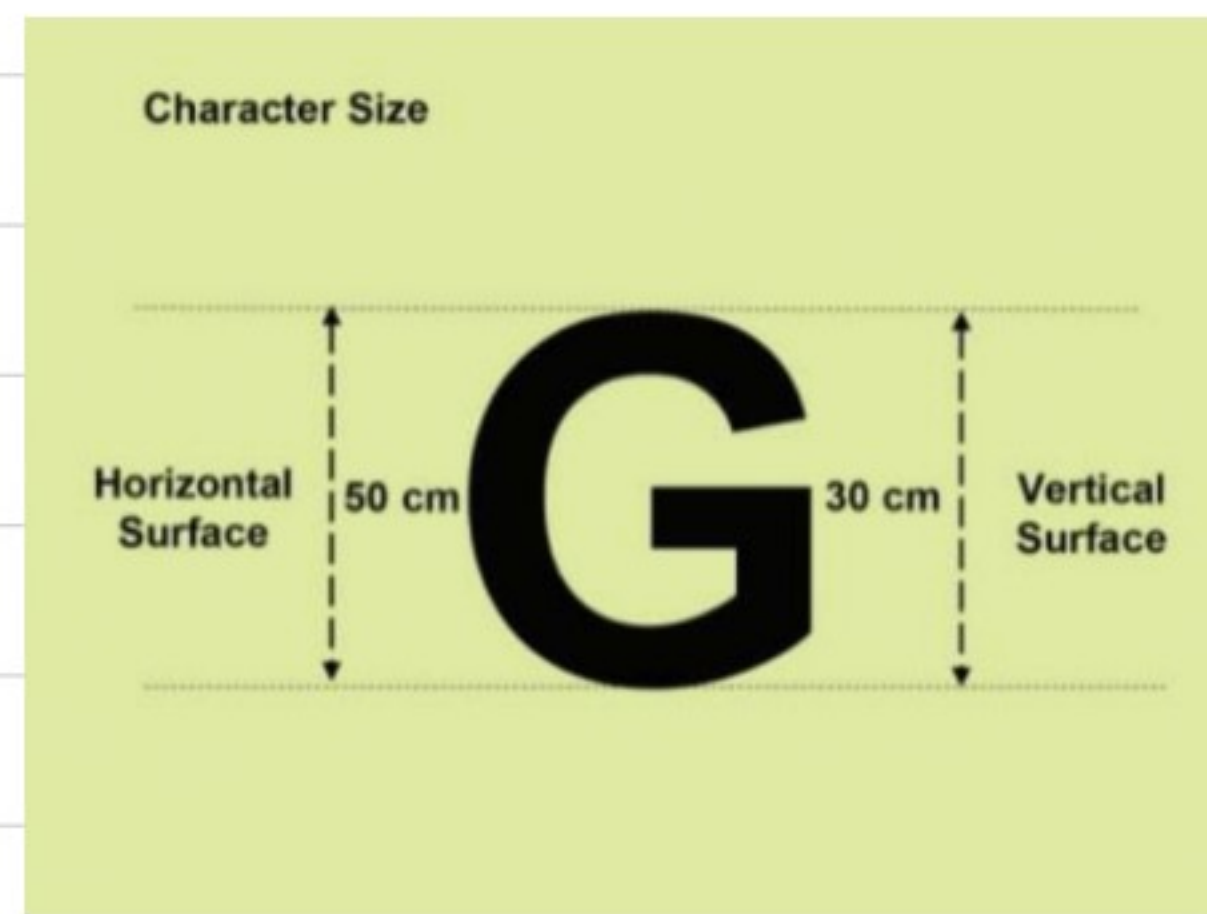
- In 1919, Rome / Paris convention, it was decided that every A/C should have a nationality and registration mark.
- Registration mark of Indian aircrafts -
  - VT - Civil A/C
  - VU - Military A/C
  - U - Drone
- Nationality Mark comprises of
  - Alphabet - C - Canada
  - Alphabet & Alphabet - VT - India
  - Number & Alphabet - 9M - Malaysia
- Registration Mark
  - Alphabet → VT - CCA
  - Numeric → VT - 7376
  - Alpha Numeric → VT - 493B



- International Telecommunications Union (ITU) issues each state a nationality mark.
- India was given 'AT' or 'AW', but India still uses VT as it was already in use.

### • Registration Mark Size

- On fuselage towards the tail - 30 cm
- Below Wing - 50 cm
- On lighter than air balloon - 50 cm



- Registration Marks **cannot** be following distress codes-

XXX - Urgency

PAN - Urgency

TTT - Security / Safety

SOS - Save our souls (Distress)

- Registration Marks also **cannot** be Q codes

QNH

QFE

QVJ

QTE

QDR, etc.

- Common Marks - They are issued to operators operating as local carrier in **more than 2 states**.  
It can be 4YA to 4YZ

# Air Traffic Services

## Annex-8

Why do we have Air Traffic Services -

- 1) Obstacle clearance
- 2) Collision Avoidance
- 3) Flow of Traffic
- 4) Advice for safe & efficient conduct of flight
- 5) Sequencing
- 6) Search & Rescue

## FIR's in India

There are 4 main FIR's in India -

- Delhi FIR (North) - I - VIDP
- Calcutta FIR (East) - E - VECC
- Bombay FIR (West) - A - VABB
- Madras FIR (South) - O - VOMM

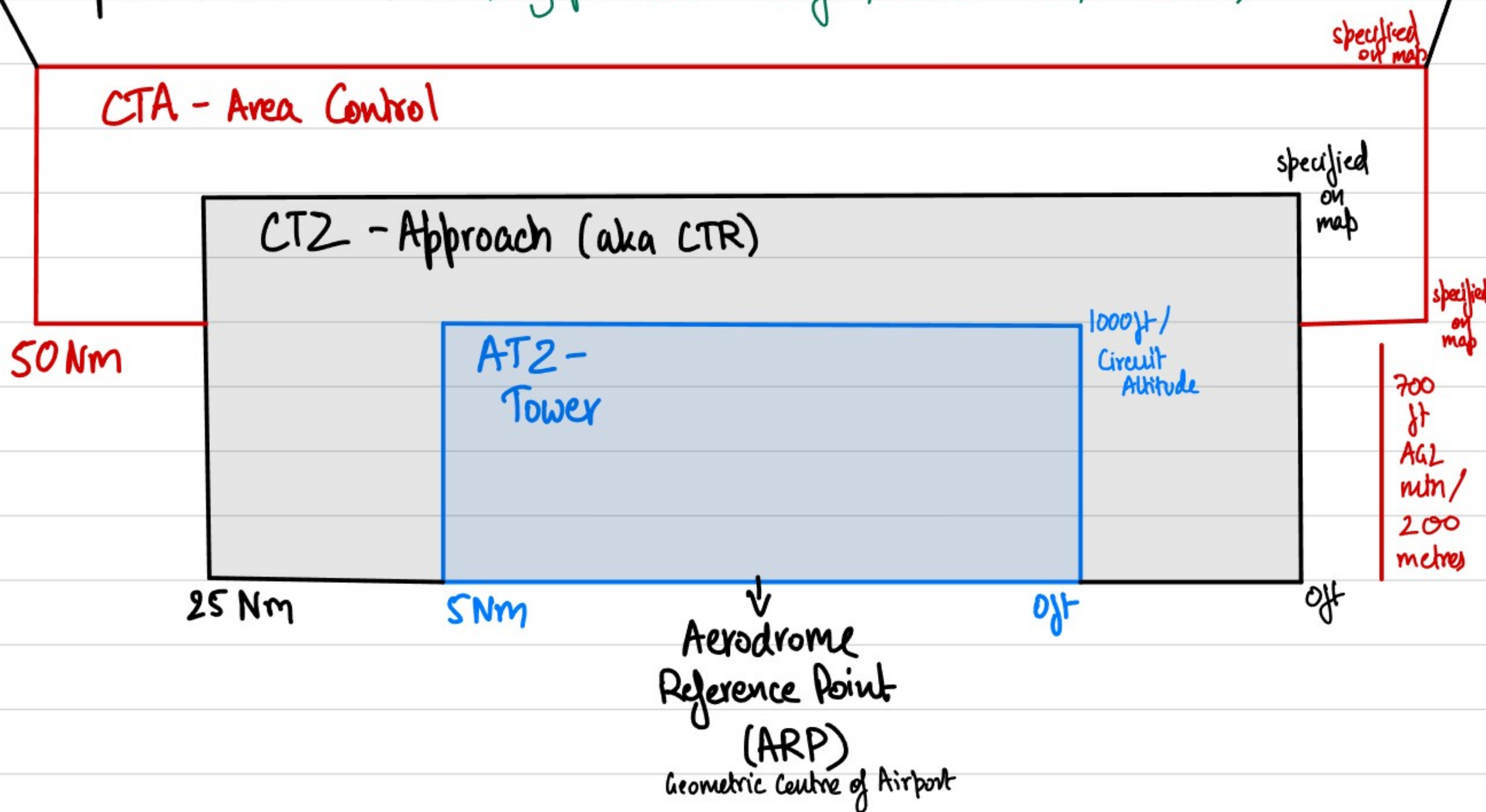
East was difficult to manage, so 1 sub FIR was introduced

- Guwahati FIR (East Sub) - VEGT

\* NOTE - The second letter in ICAO airport name tells us what FIR it is under.

Further divided in Control Areas (CTA) & Control Zones (CTZ)

Open FIR → FIS (They provide critical info, such as QNE, QNH, etc)



- Control Zone can have more than 1 Air Traffic Zone.
- Control Area can have more than 1 CTZ.
- Open Area is less regulated.

ATZ -

Vertical : 0 ft - 1000 ft / Circuit Altitude

Lateral : 5 Nm

Controlling Authority : Tower (Provides take off and landing clearances and guides on ground movement on some aerodromes.)

CTZ / CTR -

Vertical : 0 ft - Specified

Lateral : 25 Nm

Controlling Authority : Approach (Does sequencing for arriving & departing aircraft)

CTA -

Vertical : specified - specified

Lateral : 50 Nm

Controlling Authority : Area Control (Route Establishment)

\* NOTE - Approach & Area Control provide services with the help of radar. Therefore, there may be cases these 2 will be referred to as radar.

## IFR - VFR

### Instrument Flight Rules

- Flying using flight instruments as a reference.
- Can be flown in most weather conditions
- Bad weather is known as IMC

Instrument Meteorological Conditions

### Visual Flight Rules

- Flying using ground features as a reference.
- Can only be flown in good weather.
- Good weather is referred to as VMC

Visual Meteorological Conditions

To solve the dilemma caused by the word good weather, there was a need to specify a standard.

These standards were called VMC Met Minimas

C G S  
1 5 1 5  
1 5 1 5  
1 5 1 8

C G S  
1 5 1 5  
1 5 1 5  
1 5 1 8

C G S  
1 5 1 5  
1 5 1 5  
1 5 1 8

1518

- 1500 m horizontal from clouds
- 1000ft below the clouds (vertical)
- 8 km visibility



10000ft /  
FL100

1515

- 1500 m horizontal from clouds
- 1000ft below the clouds (vertical)
- 5 km visibility



3000ft QNH  
1000ft AGL

Controlled  
(with ATC)

Uncontrolled  
(without ATC)

1515

- 1500 m horizontal from clouds
- 1000ft below the clouds (vertical)
- 5 km visibility

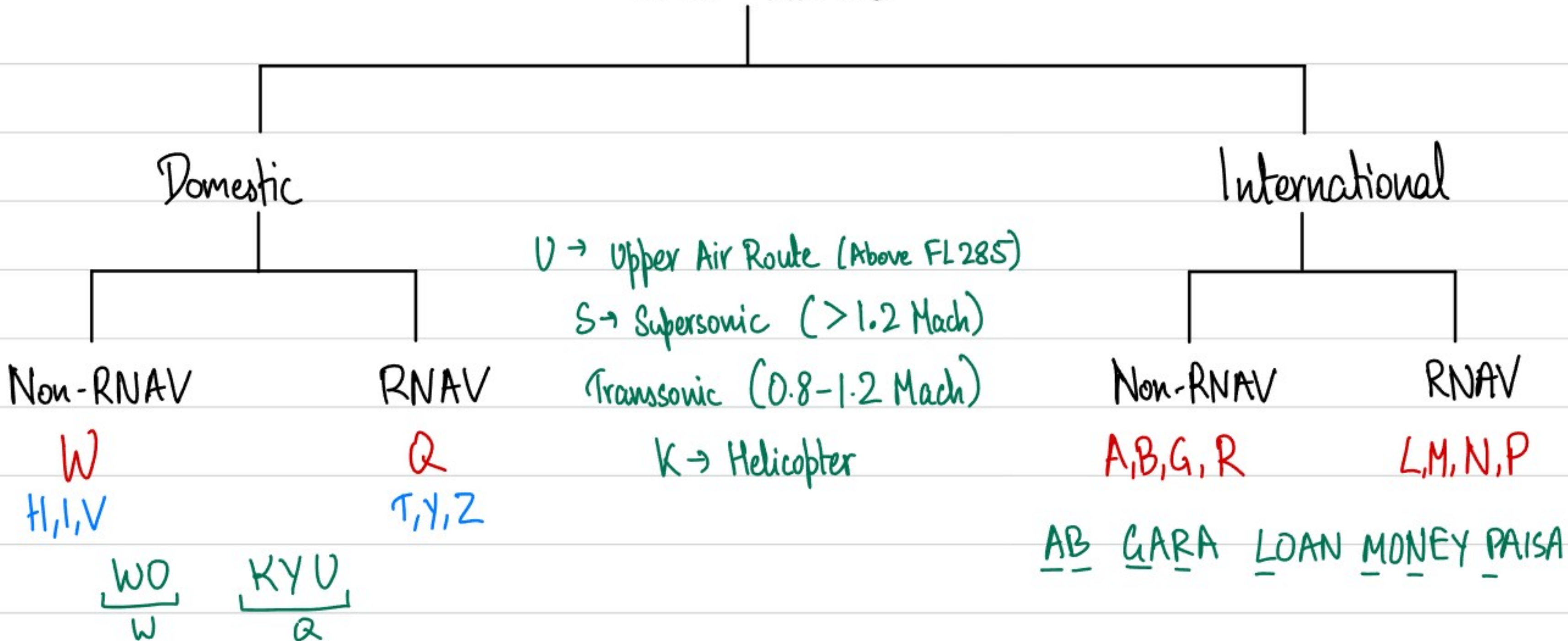
CG5

- Clear of clouds
- Ground visible
- 5 km visibility

- When flying within a controlled aerodrome within a controlled zone, the 5km visibility can be reduced to 3km if the aircraft is flying at slow speeds and there is less local traffic.

ATS Routes-

## ATS Routes



# ATS Airspace Classifications

| Class | Type of Flight | Separation Provided                      | Service Provided                                                                                          | Speed Limitation                | Radio Comms.     | Subject to ATC clearance |
|-------|----------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------|------------------|--------------------------|
| A     | IFR            | IFR-IFR<br>IFR-VFR<br>VFR-IFR<br>VFR-VFR | All air traffic control services                                                                          | Not Applicable                  | Continuous 2 way | Yes                      |
| B     | IFR            | IFR-IFR<br>IFR-VFR<br>VFR-IFR<br>VFR-VFR | All air traffic control services                                                                          | Not Applicable                  | Continuous 2 way | Yes                      |
|       | VFR            | VFR-IFR<br>VFR-VFR                       | All air traffic control services.                                                                         | Not Applicable                  | Continuous 2 way | Yes                      |
| C     | IFR            | IFR-IFR<br>IFR-VFR                       | All air traffic control services                                                                          | Not Applicable                  | Continuous 2 way | Yes                      |
|       | VFR            | VFR-IFR                                  | 1) ATS for separation from IFR<br>2) VFR traffic information services (traffic avoidance on request)      | 250 kts IAS below<br>10,000 ft. | Continuous 2 way | Yes                      |
| D     | IFR            | IFR-IFR                                  | 1) Air traffic control service<br>2) Traffic Information Service about VFR flights (avoidance on request) | 250 kts IAS below<br>10,000 ft. | Continuous 2 way | Yes                      |
|       | VFR            | -                                        | Traffic Information about IFR/VFR (traffic avoidance on request)                                          | 250 kts IAS below<br>10,000 ft. | Continuous 2 way | Yes                      |
| E     | IFR            | IFR-IFR                                  | 1) Air Traffic Control Service as far as practical<br>2) Traffic Information about VFR.                   | 250 kts IAS below<br>10,000 ft. | Continuous 2 way | Yes                      |
|       | VFR            | -                                        | Traffic Info. as far as practical                                                                         | 250 kts IAS below 10k ft.       | No               | No                       |
| F     | IFR            | IFR-IFR<br>(as far as practical)         | 1) Air Traffic Advisory Service<br>2) Flight Information Service                                          | 250 kts IAS below<br>10,000 ft. | Continuous 2 way | Yes                      |
|       | VFR            | -                                        | Flight Information Service                                                                                | 250 kts IAS below 10k ft.       | No               | No                       |
| G     | IFR            | -                                        | Flight Information Service                                                                                | 250 kts IAS below 10k ft.       | Continuous 2 way | No                       |
|       | VFR            | -                                        | Flight Information Service                                                                                | 250 kts IAS below 10k ft.       | No               | No                       |

\* NOTE - 1) In class D, IFR + VFR need ATC clearance, although ATC are responsible for IFR-IFR separation only.

2) In class E, ATC does not provide IFR-VFR separation, IFR shares responsibility for separation from VFR.

3) Class E airspaces cannot be used as control zones (CTZ).

4) Class F is known as advisory airspace.

## VMC Minimas\* (with respect to Airspace)

|    | Altitude Band                                                                                                               | Airspace Class  | Flight Visibility | Distance for cloud                                                                         |
|----|-----------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|--------------------------------------------------------------------------------------------|
| 1) | At and above 3,050 m<br>10,000 ft. AMSL                                                                                     | A***<br>BCDEFG  | 8 km              | 1500 m horizontally<br>300m (1000 ft) vertically                                           |
| 2) | Below 3050m (10,000ft) AMSL and<br>above 900m (3,000ft) AMSL or<br>above 300m (1000ft) above terrain<br>whichever is higher | A***<br>BCDEFG  | 5 km              | 1500 m horizontally<br>300m (1000 ft) vertically                                           |
| 3) | At and below 900m (3000 ft) AMSL<br>or 300 m (1000 ft) above terrain,<br>whichever is higher                                | A*** BCDE<br>FG | 5 km<br>5 km**    | 1500 m horizontally<br>300m (1000 ft) vertically<br>Clear of clouds with surface in sight. |

\* NOTE- \* When the height of transition altitude is lower than 3050 m (10,000 ft) AMSL, FL100 should be used in lieu of 10,000 ft

\*\* When so prescribed by the appropriate ATS authority,

- flight visibilities reduced to not less than 3000 m may be permitted for flights operating -

- 1) At speeds that, in prevailing visibility, will give adequate opportunity to observe other traffic or obstacles in time to avoid collision i.e. Low speeds, or

- 2) In circumstances in which probability of encounters would be low i.e. low traffic

- Helicopters may be permitted to operate upto 1500 m visibility, if manoeuvred at low speeds.

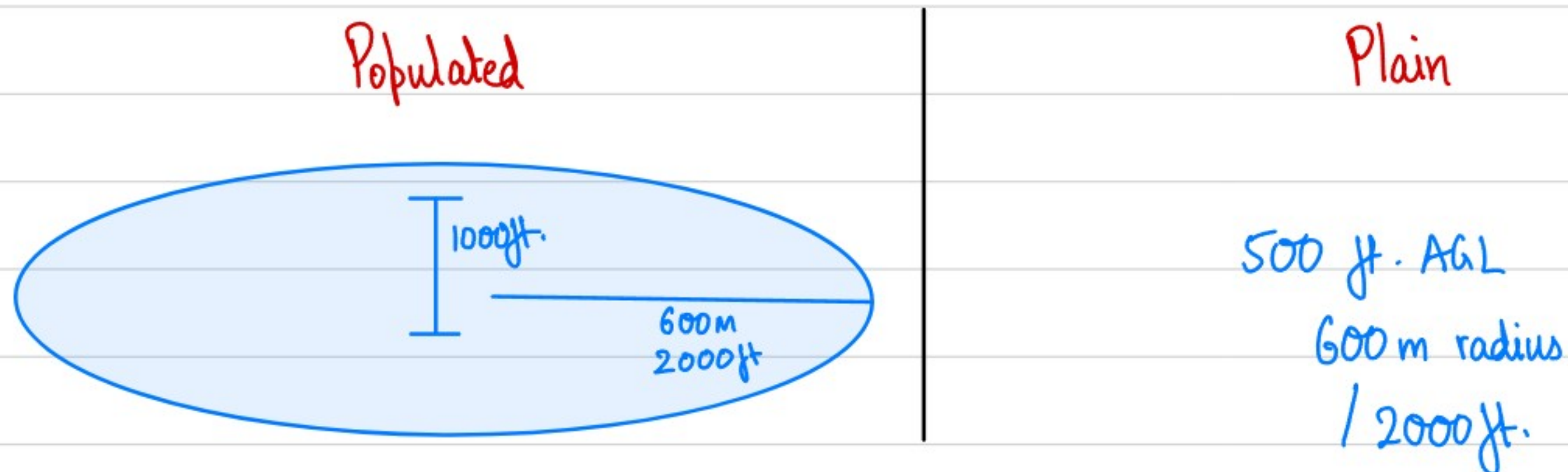
\*\*\* The VMC minima in Class A airspace are included for guidance to pilots and do not imply acceptance of VFR flights in class A airspace.

## IFR/VFR Rules

VFR flights are not authorized in the following cases -

- At night time
- At or above FL150
- More than 100 Nms seaward
- At transonic (0.8-1.2 Mach) and supersonic (Mach 1.2 and above) speeds.
- VFR cannot operate in ATZ or CTZ without permission in less than 5km visibility and 450m/1500ft ceiling.
- Night Time →
  - 20 minutes after sunset.
  - 20 minutes before sunrise.
- Night Time can be logged →
  - 30 minutes after sunset.
  - 30 minutes before sunrise.

Minimum heights to fly under VFR. unless SVFR



Minimum heights to fly under IFR unless following 'MEA' → Route Guide.

| Mountaneous                        | Plain                              |
|------------------------------------|------------------------------------|
| 2000ft AGL<br>5Nm radius<br>/ 8 km | 1000ft AGL<br>5Nm radius<br>/ 8 km |

## Provisions for IFR -

1) Equipment - IFR flights should have suitable instruments and navigation aids.

2) Levels to be maintained - • Minimum Enroute Altitude • From operations manual if published

• Else, 2000 ft. above obstacle in hilly areas } 5 Nm  
or 1000 ft. above obstacles in plain areas } 8 km

• Above the minimums, A/C shall maintain altitude as per semi-circular rule.

3) A/C under IFR shall inform ATC if,

1) TAS changes by  $\pm 5\%$

2) ETA changes by  $\pm 2$  minutes

3) Change in route or destination.

\* 4) Reporting Position -



Night

30 minutes always

5) A/C shall not change from IFR to VFR without ATC permission.

\* 6) ADC / FIR number needs to be reported and permission is needed when crossing over an FIR.

## Provisions for Special VFR

1) Special VFR permission granted to A/C and pilot must hold IR/AFIR/FIR.

2) Permission granted by Approach Control or higher agency.

3) SVFR may be permitted to operate in controlled airspace if visibility is not less than 1500m and adequate arrangements have been made in advance.

Note: AIRREP (Air Report)

Section 1: Position Report

• Identification

• Position / Time etc.

Section 2: Operational Information

• ETA and endurance

Section 3: PIREP (Pilot's Report)

• Meteorological Information

## Prohibited, Restricted & Dangerous Areas.

### Airport Name Nomenclature



Prohibited (P)

Restricted (R)

Dangerous (D)

ex.



VIR

VOP

it may have a no. after it  
never allowed to fly here

can be allowed with permission

allowed but with extreme caution

- 1) Prohibited Area -
- An airspace of defined dimensions above the land area or territorial waters of a state within which the flight of an aircraft is totally prohibited.
  - You cannot enter prohibited area even with ATC permission.

- 2) Restricted Area -
- An airspace of defined dimensions above the land areas or territorial waters of a state within which the flight of an aircraft is restricted in accordance with certain specified conditions.
  - You can enter restricted area by taking ATC permission and following the restrictions.

- 3) Danger Area -
- An airspace of defined dimensions above the land areas within which dangerous activity to the flight of aircraft may exist at certain times.
  - You can enter without ATC permission but take precautions while flying.

Steps to be taken when you enter any of the above airspace (when you are not supposed to) -

- 1) **Identify**: A series of projectiles discharged at intervals of 10 seconds showing or bursting red and green lights.
- 2) **Indicate**: The visual signal indicates that the A/C is flying in vicinity of restricted, prohibited, or danger area and should take necessary actions.
- 3) **Necessary Actions**:
  - Do not fly further into the area.
  - Do not descend while still above the area.
  - As soon as we are aware of the fact, give signal of distress - MAYDAY, MAYDAY, MAYDAY by RT.
  - Immediately squawk distress code 7700 and land at nearest aerodrome outside prohibited area.

| Name                                    | Restriction Area                    |
|-----------------------------------------|-------------------------------------|
| Tower of Silence, Malabar Hills, Bombay | 1 Nm Radius                         |
| Baroda Aerodrome                        | Ground to Unlimited Upper Level     |
| Rashtrapati Bhavan                      | Ground to Unlimited Upper Level     |
| Mathura Refineries                      | Radius of 10 km                     |
| Area Near Bhubaneswar                   | Ground Level to 50,000 ft.          |
| Kalpakkan Nuclear Installation          | Radius - 10 km, Ground - 10,000 ft. |

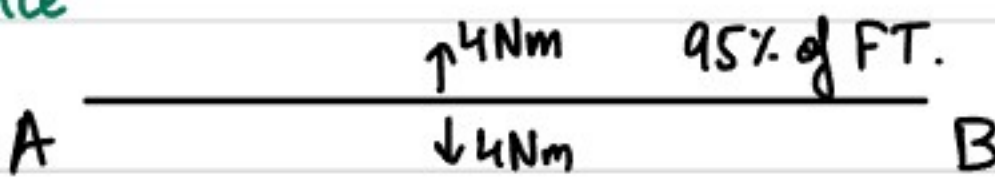
## ATIS - Aerodrome Terminal Information Service

- It is NOT a weather forecast.
- It is a routine information which is broadcasted repeatedly about aerodrome (RWY in use, winds, approach etc.)
- ATIS broadcast shall only be transmitted on VOR frequency but not on voice channel of ILS ever.
- ATIS is provided, i.e. preparation and cancellation of ATIS message, is the responsibility of air traffic service.
- ATIS broadcast message should whenever practicable not exceed 30 seconds.
- ATIS broadcast message containing arrival and departure information should include cloud cover below 5000 ft or below the minimum sector altitude whichever is greater.
- ATIS information is updated when there is any official weather regardless of the reported values.

RCP - Radio Communication Performance

- Ex. RCP<sub>10</sub> - where 10 is the no. of seconds it takes for an instruction to travel from ground to aircraft and acknowledgement back to the ground.

RNP - Radio Navigation Performance

- Ex. RNP<sub>4</sub> - where 4 is Nm. 
- The navigation accuracy of  $\pm 4$  Nm on 95% of the flight time. (for ex. above)

RSP - Required Surveillance Performance

- Ex RSP<sub>180</sub> - it is time in seconds (180) between flight crew / HMI and controller/HMI.

RNAV - Area Navigation

## Aircraft Emergencies

- An air **accident** is required to be reported to DGCA within 24 hours.
- An accident is said to have occurred when-
  - 1) A person in the aircraft suffers death or serious injury.
  - 2) The aircraft suffers substantial damage.
- The wreckage may not be removed without DGCA permission unless it is blocking a runway or causing obstruction
- All other events that affect the aircraft are labelled **incidents**.

**Ditching** - The forced landing of an A/C on water.

**Air Miss / Air Prox** - It is a report raised by a pilot if he sights an A/C that is dangerously close.

## Emergencies

- 1) **Distress** - • An aircraft facing imminent danger and requiring immediate assistance.
  - They will announce **MAYDAY** three times.
- 2) **Urgency** - • An aircraft is in a situation where there is no imminent danger but could turn into an emergency if assistance is not provided.
  - They will announce **PAN-PAN** three times.

## Distress Frequencies aka Guard Frequencies

- 1) **121.5 Mhz / 243 Mhz** - International VHF distress frequencies.
- 2) **2182 / 8364 khz** - International MF distress frequencies.
- 3) **500 khz** - International Maritime distress frequency.

## Search and Rescue Organisation (SAR)

- 1) Joint Rescue Coordination Centre (JRCC) - Responsible for both aeronautical and maritime SAR.
- 2) Rescue Coordination Centre (RCC) - Conducts SAR within a particular region.
- 3) Search and Rescue Region (SRR) - Have defined areas and are linked to RCC. Boundaries same as FIR.
- 4) Rescue Subcentre (RSC) - Also comes under RCC.
- 5) Alerting Post - Any facility between person reporting & RSC. Ex. Police.
- 6) Search and Rescue Unit / SAR Mobile Team - Trained personnel that physically carry out the operation.

## Phases of Emergency

- 1) INCERFA - Uncertainty about the safety of an A/C and its occupants. Ex. No radio contact or A/C overdue by more than 30 mins of its ETA.
- 2) ALERFA - Alert Phase. When there is apprehension about the safety. Ex. A/C in circuit fails to land within 5 minutes of its expected landing time.
- 3) DITRESFA - Distress Phase. When there is reasonable certainty about the A/C being in distress. Ex. An A/C carrying 5 hours of fuel has not landed for more than that time.

Action by Search A/C - When search A/C spots survivors and observes signals -

- 1) Rocks wings during daytime.
- 2) Switching landing / navigation lights on/off twice.

Emergency Locator Transmitter (ELT) - is equipped in A/C and transmits distress message automatically on 121.5 / 406 MHz if crash creates an impact of 5-7 g or more. Batteries last 48 hours.

ELT transmissions are picked up by SARSAT satellites and passed on to local transmitters at Bangalore and Lucknow from where they are passed on to the 4 FIR's.

Dropped Emergency Packages - Red packages contain first aid.

Green packages contain food and beverages.

Black packages contain miscellaneous items such as clothes etc.

## Ground Air Visual Codes by survivors

| No. | Message                      | Code Symbol |
|-----|------------------------------|-------------|
| 1   | Require Assistance           | ✓           |
| 2   | Require Medical Assistance   | ✗           |
| 3   | No or Negative               | N           |
| 4   | Yes or Affirmative           | Y           |
| 5   | Proceeding in this direction | ↑           |

## Ground Air Visual Codes by Rescue Units

| No. | Message                                               | Code Symbol                                                                           |
|-----|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1   | Operation Completed                                   | LLL                                                                                   |
| 2   | We have found all personnel                           | <u>LL</u>                                                                             |
| 3   | We have found only some personnel                     | ++                                                                                    |
| 4   | We are not able to continue. Returning to base.       | XX                                                                                    |
| 5   | We have divided into groups. Proceeding in indicated. |  |
| 6   | Information Received. A/C in this direction.          |  |
| 7   | Nothing found. Will continue to search.               | NN                                                                                    |

# Communication

## Commonly used Q Codes -

- 1) QDL - Series of bearings
- 2) QDM - Magnetic heading towards.
- 3) QDR - Magnetic heading from.
- 4) QFE - Altimeter subscale setting for height above reference.
- 5) QFU - Runway Magnetic Orientation.
- 6) QGH - Type of approach for landing.
- 7) QNE - Altimeter indication at standard pressure.
- 8) QNH - Altimeter subscale setting for elevation
- 9) QTE - True bearing from
- 10) QVJ - True bearing to

In case IMC conditions  
continue

## Categories of messages (with priority level)

- 1) Distress (Mayday)
- 2) Urgency (Pan Pan)
- 3) Direction Finding
- 4) Flight Safety
- 5) Meteorological Messages
- 6) Flight Regularity Messages.

\* NOTE - Check Pg 39, 40, 41 for standard words/phrases and pronunciation.

## Radio Check Scale (How do you read)

- 1) 1 - Unreadable
- 2) 2 - Readable now and then
- 3) 3 - Readable but with difficulty
- 4) 4 - Readable
- 5) 5 - Perfectly Readable

# Personnel Licensing

## Student Pilot's License (SPL)

### Requirements

- 1) Age - 16 y.o.
- 2) Medical - Class 2
- 3) Examination - Oral Examination - Air Regs, Meteorology, Air Navigation, Technical Specific.

### Privileges

- 1) We shall fly at all times under supervision of a flight instructor or examiner.
- 2) We shall fly under VFR only.
- 3) We cannot carry passengers.
- 4) We can carry out cross country with minimum of 10 hours of solo time.

## Private Pilot's License (PPL)

### Requirements

- 1) Age - 17 y.o.
- 2) Medical - Class 2
- 3) Examination - Meteorology, Navigation (Confirm Once)

### Privileges

- 1) Can carry passengers but not for remuneration.

## Commercial Pilot's License (CPL)

### Requirements

- 1) Age - 18 y.o.
- 2) Medical - Class 1
- 3) Examination - Air Regs, Met, Navigation, Tech Gen, Tech Specific, RTR.

### Privileges

- 1) Can fly for remuneration
- 2) Can act as PIC of A/C upto 5700 kgs.

## CPL

- 1) Minimum 200 hours
- 2) 100 hrs PIC
- 3) Min. 20 hours of X-Country solo
  - 300 Nm (X-Ctry with 2 technical landings)
  - 250 Nm (Nav check)
- 4) Min 10 hrs instrument time (5 hrs can be on sim)
- 5) 5 hrs night solo (10 T/O, 10 Landings)
- 6) Daytime / Nighttime check (DT/NT)

## CPL + IR

- 1) Minimum 200 hours
- 2) 100 hrs PIC
- 3) Min. 50 hours of X-Country solo
  - 300 Nm (X-Ctry with 2 technical landings)
  - 250 Nm (Nav check)
- 4) Min 40 hrs instrument time (20 hrs can be on sim)
- 5) 5 hrs night solo (10 T/O, 10 Landings)
- 6) Daytime / Nighttime check (DT/NT)
- 7) IR check

## Airline Transport Pilots License (ATPL)

### Requirements

- 1) Age - 21 y.o.
- 2) Medical - Class 1
- 3) Examination - General, Radio, Meteorology (Written) (note: Gen & Radio have an oral exam as well. They are valid for 2.5 years without clearing oral. 5 after.)

### Flying Requirements

- 1) Min 1500 hours
  - 2) Min 1000 hours of cross country
  - 3) Min 100 hours night time
  - 4) Min 100 hours instrument time.
  - 5) Min 500 hours solo PIC.
  - 6) Min 200 hours cross country solo.
  - 7) Min 50 hours cross country night.
- } includes co-pilot + PIC.  
note: not more than 50% can be logged as co-pilot time  
the rest will be PIC. An examiner can give full PIC or co-pilot.

### Privileges

- 1) Can act as PIC of aircraft over 5700 kgs.

## Flight Plan

• Flight Plan is needed for the following reasons-

- 1) Any flight or portion of flight requiring ATC service.
- 2) Any IFR flight in advisory airspace.
- 3) Any flight where appropriate ATC service is required to provide flight information service, alerting service, search and rescue service.
- 4) Any flight across international borders.
- 5) Where the authority has determined that a flight plan should be filed to facilitate coordination between civilian and military authorities.

### \* NOTE -

- 1) For a flight to be provided with ATC services or an advisory ATC, the flight plan is to be filed **at least 60 minutes** before departure.
- 2) For filing a flight plan in flight, it is to be filed 10 minutes before aircraft is estimated to reach -
  - a) The intended point of entry into an area where ATC advisory is to be provided.
  - b) The point of crossing an airway or advisory route.

### Delays After Filing

- In an event of delay of 30 minutes for a controlled flight OR a delay of 1 hr for an uncontrolled flight for which flight plan has been submitted, the flight plan should be amended or a new flight plan submitted and the old flight plan cancelled.

\* NOTE - ADC number is valid from -15 mins to +45 mins from filed EOBT (Estimate of Block Time)

### Types of Flight Plans

- 1) **FPL** - Filed flight plan
- 2) **CPL** - Current flight plan
- 3) **RPL** - Repetitive flight plan. 10 flights should be flown for 10 consecutive occasions to call FPL as RPL.

\* Normally flight plan should be 60 mins before EOBT. But for AFTM (air flow traffic mgmt) file flight plan **3 hours before**.

























































































































































































